Q.2] Air Traffic Control System – AVL Tree Analysis

1. Problem

- The AVL tree is initially keyed by aircraft transponder code.
- Searching by transponder code is fast, but it does not help find the nearest aircraft.
- Changing the key to aircraft distance makes spatial searching easier.
- However, aircraft distance changes continuously.
- Every distance update requires deletion + reinsertion into the AVL tree.

2. Performance Issue

- Number of aircraft = 500
- Each aircraft updates every 2 seconds.
- Each update requires approximately 2 AVL rebalancing operations.
- This produces about 500 × 2 = 1000 rebalancing operations per second.
- This consumes around 30% CPU time, making the system inefficient.

3. Better Data Structure

A better solution is to use a spatial data structure, such as a k-d tree (k-dimensional tree).

- Store aircraft using their X and Y coordinates.
- Nearest-aircraft queries can be performed efficiently.
- Spatial regions can be searched without ordering everything by constantly changing distance.
- Altitude can be included as an additional dimension or maintained in a separate index.
- Position updates can be handled more efficiently than repeatedly modifying an AVL tree keyed by distance.

4. Proposed System

Aircraft Sensors

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Position + Altitude + Speed + Heading

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Spatial Data Structure (k-d Tree)

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↓	↓	↓
Nearest Aircraft	Altitude Range Search	Collision Detection
↓	↓	↓
Control	Tower	Alert

5. Conclusion

The AVL tree is efficient for static or slowly changing ordered data, but it is not ideal for aircraft positions that change every 2 seconds. A k-d tree or another spatial indexing structure is more appropriate because the primary operations are nearest-neighbor and spatial-range queries, not searches by transponder code.